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AI FOR SAFER ROADS INNOVATION CHALLENGE 2026

Description of data sources

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1 Data sources

1.1 Open Data

The data underlying the two different map outputs are broadly similar, although there are small differences between the two due to available data at the time both were created. Each contains a map of the target road network, to which various data have been derived and allocated. The target road network does not consist of all roads in the two areas of interest, it concerns only roads of a certain class. The network data has been downloaded from Overture[1] at a fixed point in time. For Thailand this network was accessed in December 2024, for Maharashtra it was May 2025. Overture versions will update each month for these networks which means the output will gradually become less contemporaneous. Overture data is made available under the Open Database Commons licence[2]. Attribution should therefore be given in the following format when displaying the network in a publication - *© OpenStreetMap contributors, Overture Maps Foundation.* Further attributions may apply depending on the data used.

The Overture data has been downloaded and then simplified to create a series of road sections for analysis and display purposes. When the Overture data is accessed there are many different road links; our approach within the project was to join links together to create sections which will be similar in nature and form recognisable routes. Sections typically form longer stretches of road between major junctions and are created programmatically, rather than manually. Some links may be very long and could pass through major intersections not detected in the data model. Others may appear quite short, especially in complex urban areas. No attempt was made to edit or rationalise the network in this programme.

Only road classes; Motorway, Trunk, Primary; and Secondary were accessed for the project. Each road section was also split according to whether it was in a rural or urban area. There are various methods of doing this; our approach was to use geodatabases provided by Daylight[3] which allowed us to identify the approximate locations of rural and urban areas. Speed limit data was derived from the commercial dataset used to analyse traffic flow and vehicle speeds and then matched back to the summarised Overture Sections. It is likely that speed limits along sections could change and this has not been captured in the analysis.

1.2 TomTom Data

For each Overture road section and attempt was made to access commercial data via the TomTom Move portal using their API. Rather than purchase data for the entire road section, samples were taken at a 10km interval. This means for a 20km section, two samples would be obtained. Traffic data was obtained in both directions of flow and averaged across all samples along the road section.

Traffic probe counts were use to obtain sample sizes which were then weighted by road length. This means that a short road section, say 5km, with the same traffic as a 10km section would achieve half the weighting. These weighted sample sizes were then used to create travel percentiles with each road section assigned to a banding. This allows us to assess which roads have the most travel and achieve the aim of mapping roads upon which 75% travel occurs.

[1] https://overturemaps.org/
[2] https://opendatacommons.org/licenses/odbl/
[3] https://daylightmap.org/

Traffic speed data was obtained at the same time in the same way by sampling the road network. The speed data was then downloaded and analysed to estimate the number of vehicles exceeding the posted speed limit, which was provided in the same data output. Once the number of vehicles exceeding the limit, and the total sample size was available this was included in the data attached to each segment. Finally, data on average speeds and 85th percentile speeds was also obtained for each road section.

1.3 Land Use

Each of the road sections have then been assigned to a land use category of either urban or rural using an international dataset (NASA Global Rural Urban Mapping Project – GRUMP), as described in the methodology report. This was accessed for Thailand in late 2024 and the split of rural versus urban is approximately three to one in terms of road length.

1.4 Final Data File Format

The data is hosted in an online dashboard and the files associated with the challenge are the backend databases that support the tool. The following table is a description of the fields in the output. Many of them are not required. The key fields are usually those relating to sample size and vehicle speeds. Much of the other data is derived from the donor network, Overture.

Field Name	Sample Data	Description
OBJECTID	9	Unique ID created by GIS software
english_ro	Surin Ring Road	English Street Name
OvertureID	9	Unique ID created by GIS software
SampleSize_avg	275650	Average sample size (if multiple TomTom datasets were collected)
RoadLength	3.7	Ignore, use Shape Length
WeightedSample	1019905	Average annual traffic weighted value by road length. Obtained from TomTom SampleSize
Percent_	1.95E-05	Ignore
Percentile	0.04299693	Which travel percentile does this road fit into? In this case the value of 0.042 = 5th percentile
SpeedLimit	90	Speed limit obtained by TomTom. Not validated
RoadClass	primary	Overture road class
NumberOverLimit	96979	Number of vehicles estimated to be over the speed limit per year based on the actual sample. As we do not know true AADF this is not the actual number on the road but a percentage estimate has been produced
MedianSpeed	80.925	Median (50th percentile) speed
F85thPercentileSpeed	97.25	85th percentiles speed
ForAnalysis	90	Ignore
ProvinceID	32	Province ID in Thailand only, not useful for this analysis
SpeedLimitFloor	90	Ignore

PercentOverLimit	0.351819336	Percentage estimate based on probe counts and speeds.
InvPercentile	0.95700307	Inverse of the traffic percentile figure, only use for dashboard filtering
AnalysisStatus	Valid	Internal flag for analysis validation
RankedPercentile	74.0577455	Allows presentation of roads by percentage traffic
StreetImageLink	103.48332214, 14.94148244,103.43913438,14.86699584	Lat / Lon for Google StreetView
LandUse	RURAL	Land use based on GRUMP data
NO_OF_Result_Segments	4	Ignore
PercentileBand	0-5%	Percentile band for dashboard
SampleSizeTotal	275650	Total sample size (if multiple TomTom datasets were collected)
Shape_Length	11643.83585	Geometric length of the shape